Ideal Transmission Line Example

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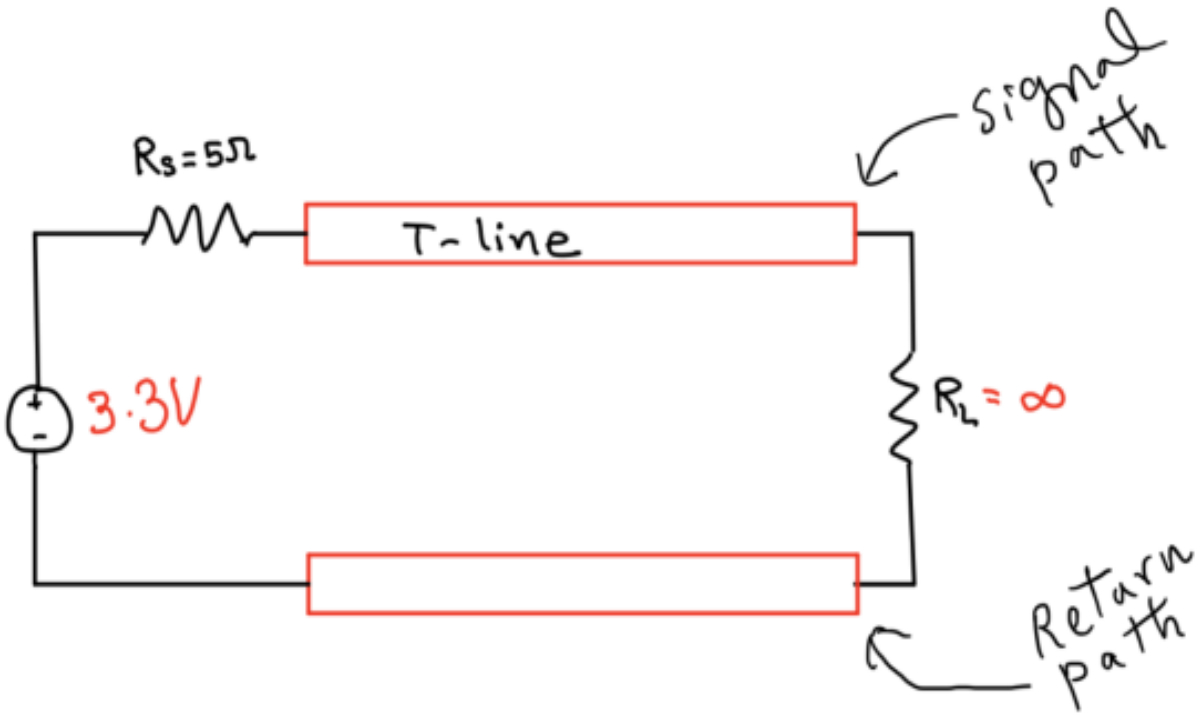


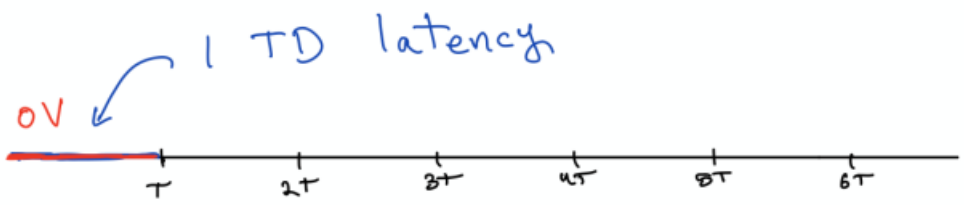
Reflection Coeficient @ Source

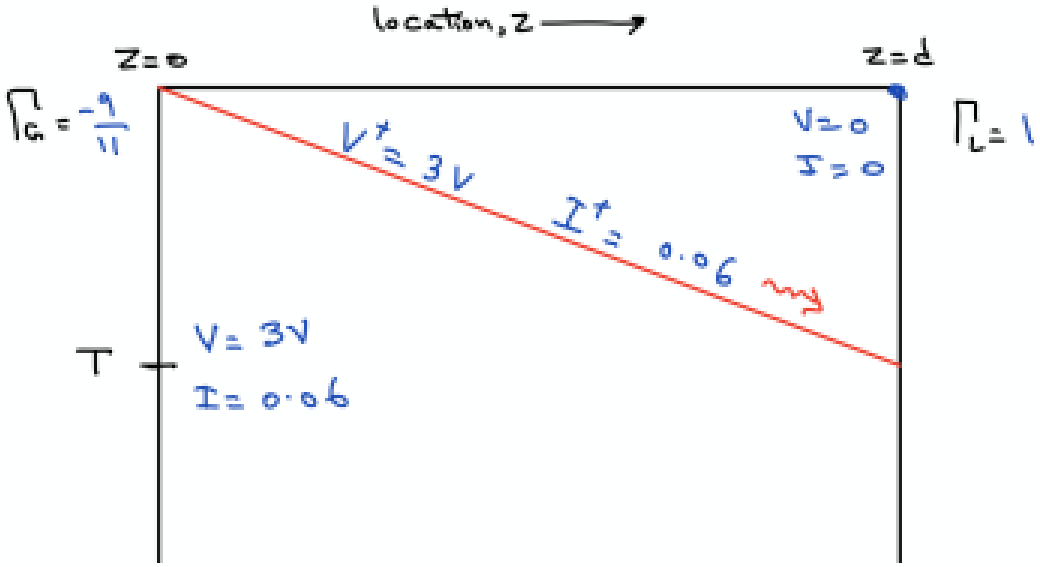
$$\Gamma_G = \frac{Z_S - Z_0}{Z_S + Z_0} = \frac{5 - 50}{5 + 50} = \frac{-9}{11}$$

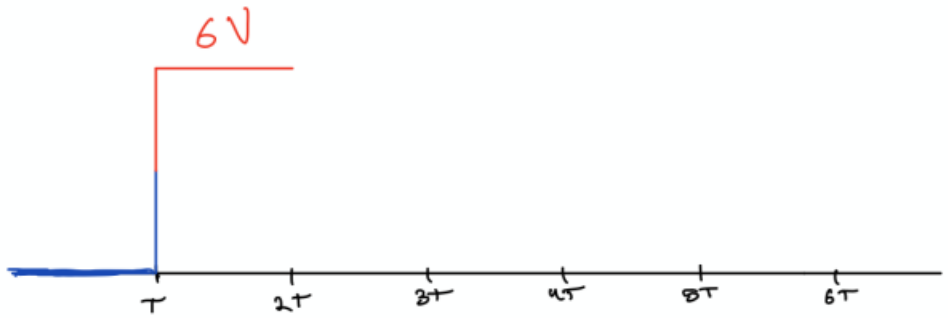
Reflection Coeficient @ Load

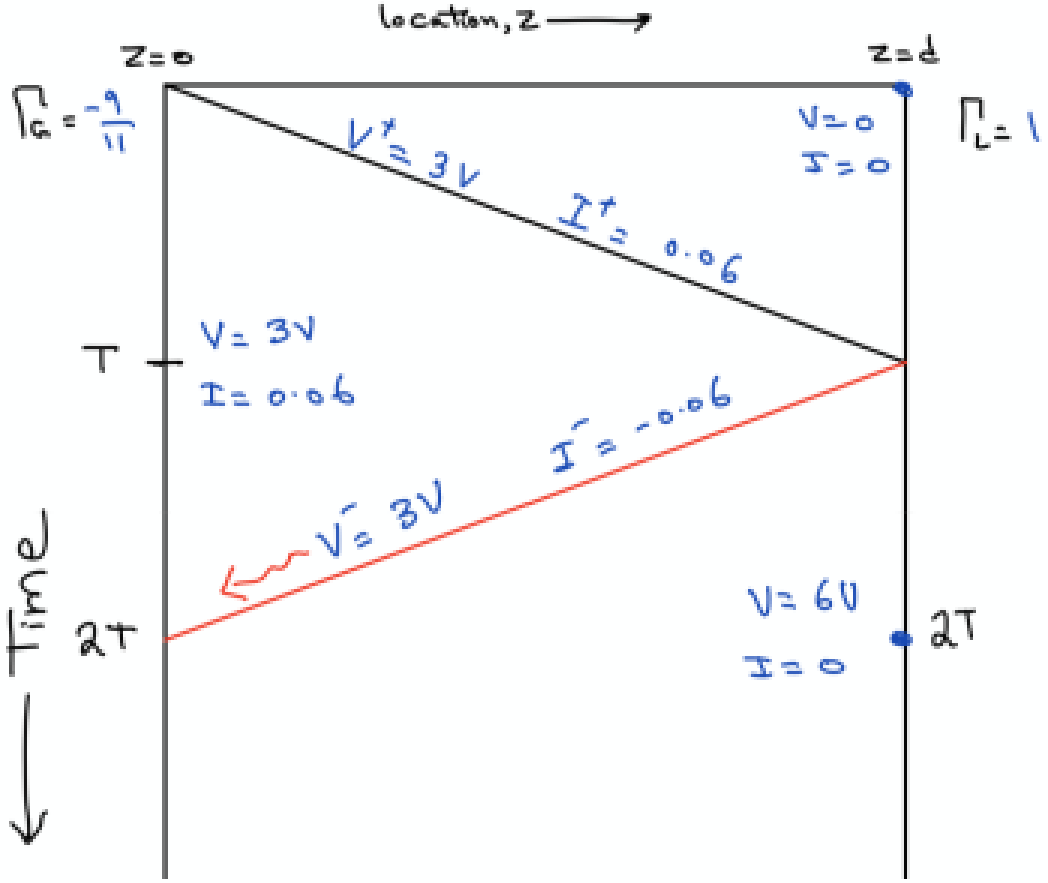
$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{1 - \frac{Z_0}{Z_L}}{1 - \frac{Z_0}{Z_L}} = \frac{1 - \frac{Z_0}{\infty}}{1 - \frac{Z_0}{\infty}} = 1$$

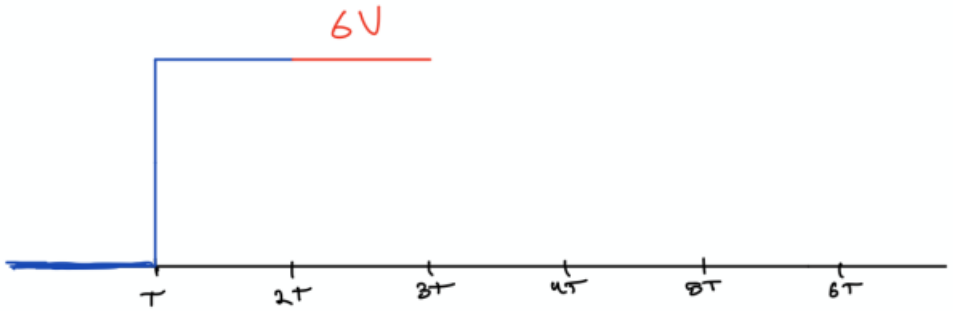
5Ω CMOS buffer driving an open ended (no load) Transmission Line with 50Ω characteristic impedance and T delay.

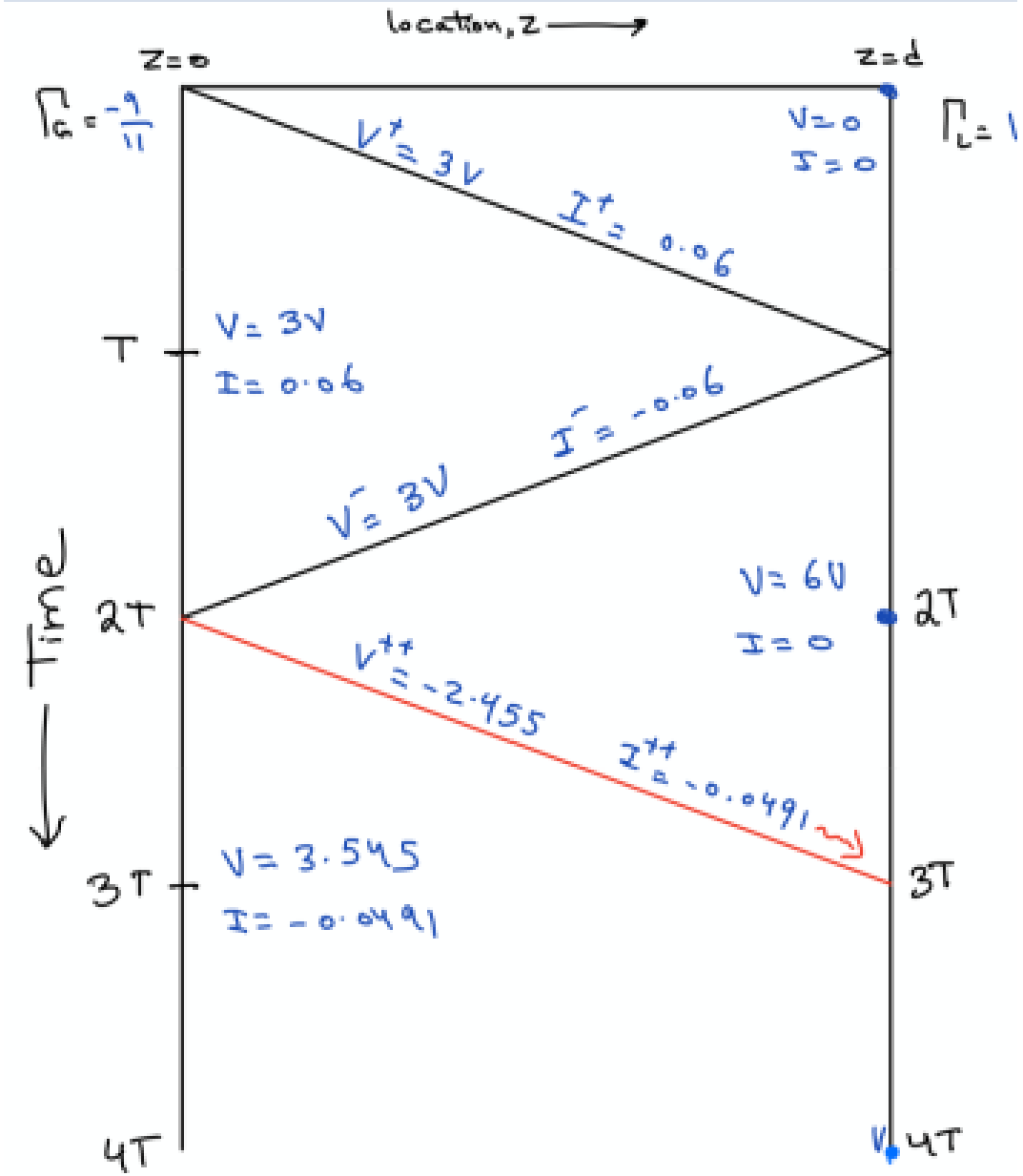

1 TD latency
0V
T
2T
3T
4T
5T
6T

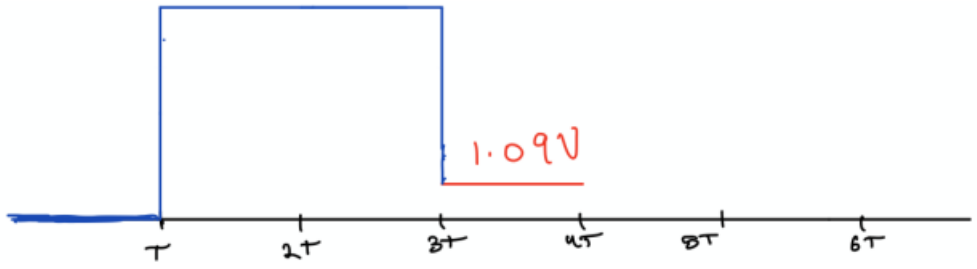

location, z
z=0
z=d
$\Gamma_G = -\frac{9}{11}$
$\Gamma_L = 1$
$V^+ = 3V$
$I^+ = 0.06$
V=0
I=0
T
V= 3V
I= 0.06

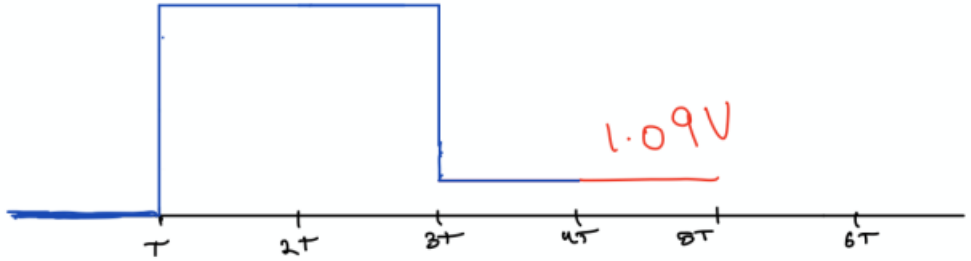

6V
T
2T
3T
4T
5T
6T

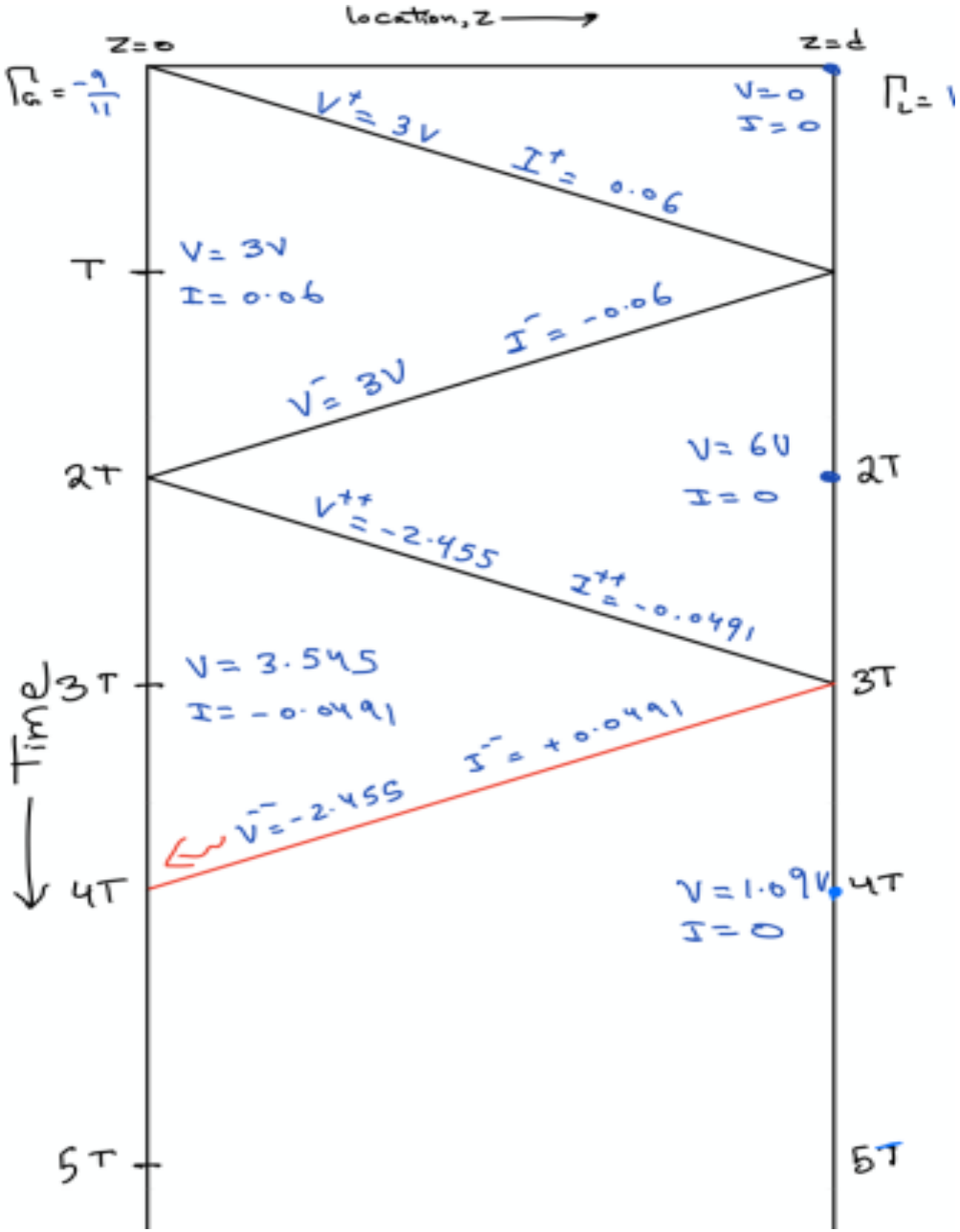

location, z
z=0
z=d
$\Gamma_G = -\frac{9}{11}$
$\Gamma_L = 1$
$V^+ = 3V$
$I^+ = 0.06$
V=0
I=0
T
V= 3V
I= 0.06
$I^- = -0.06$
$V^- = 3V$
Time
2T
V= 6V
I= 0
2T

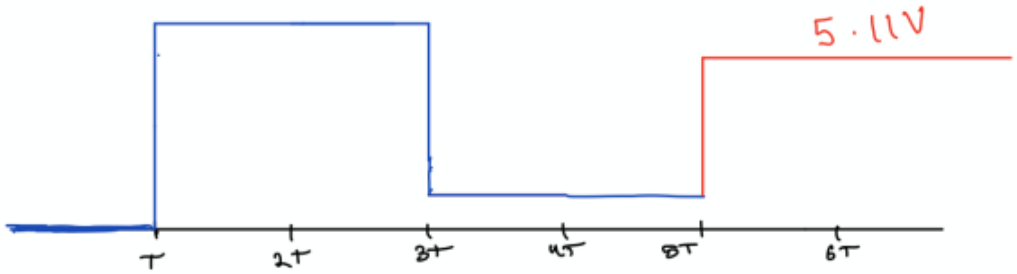

6V
T
2T
3T
4T
5T
6T

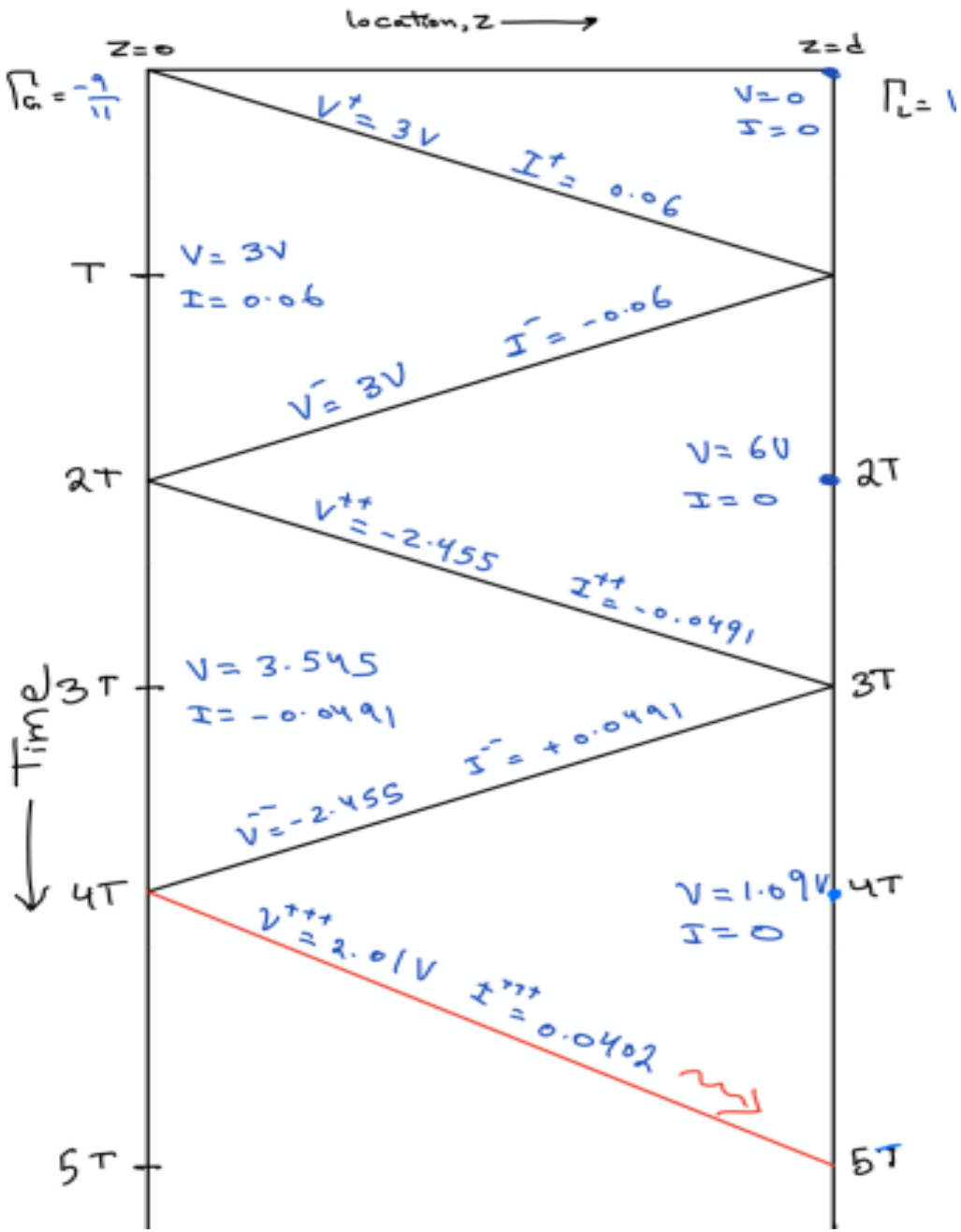

location, z
z=0
z=d
$\Gamma_G = -\frac{9}{11}$
$\Gamma_L = 1$
$V^+ = 3V$
$I^+ = 0.06$
V=0
I=0
T
V= 3V
I= 0.06
$I^- = -0.06$
$V^- = 3V$
Time
2T
V= 6V
I= 0
2T
$V^{++} = -2.455$
$I^{++} = -0.0491$
3T
V= 3.545
I= -0.0491
3T
4T
V
4T

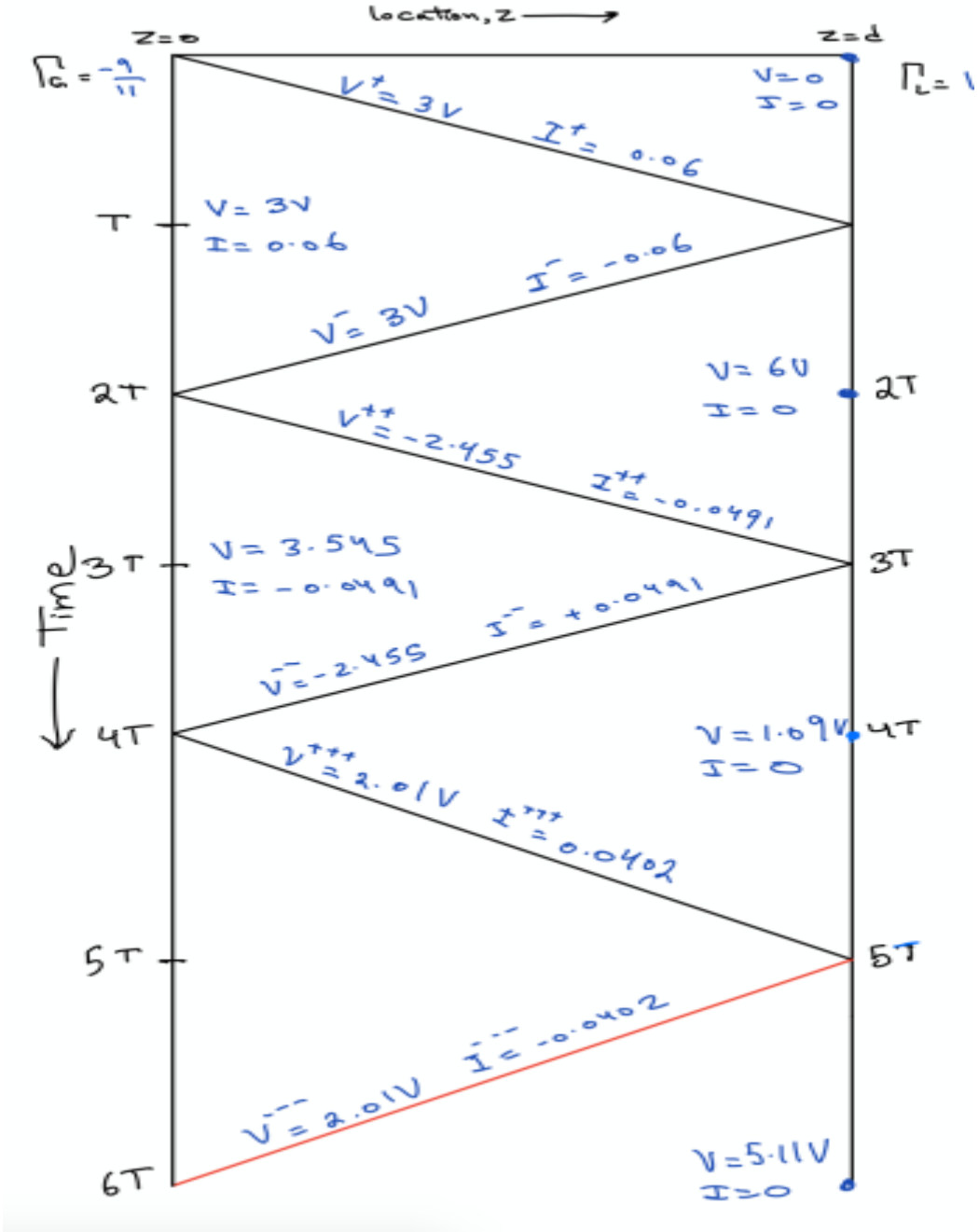

1.09V
T
2T
3T
4T
5T
6T

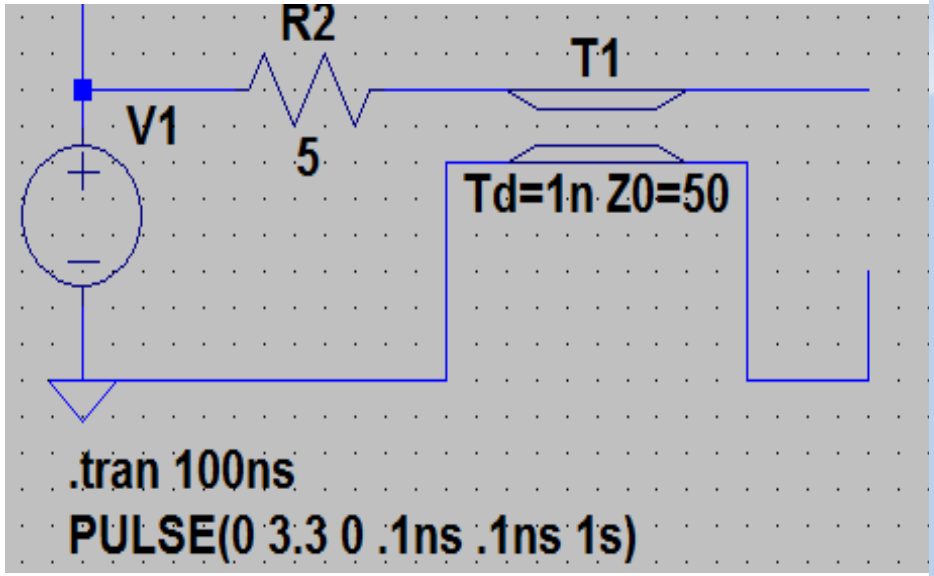

1.09V
T
2T
3T
4T
5T
6T

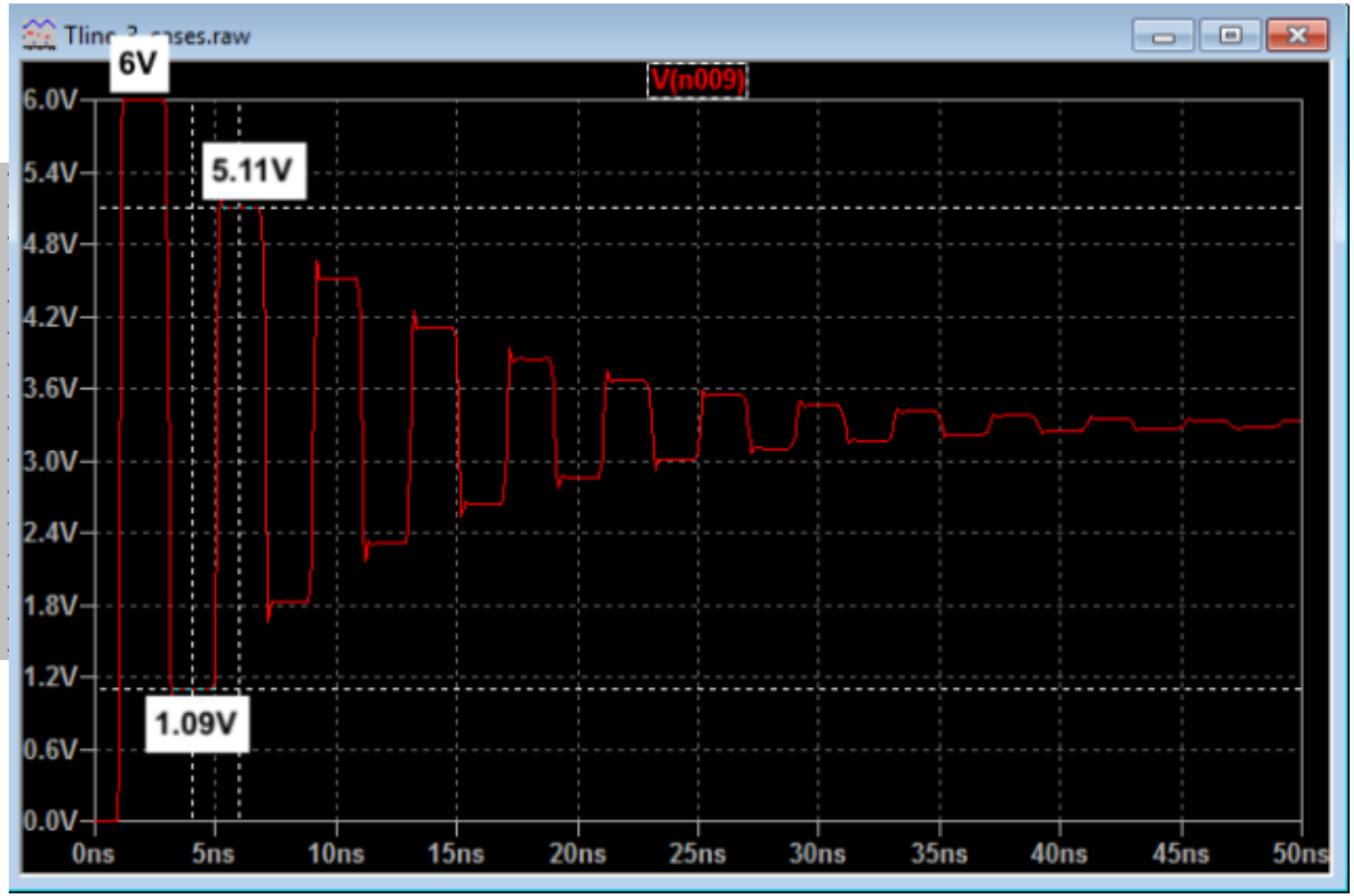

location, z
z=0
z=d
$\Gamma_G = -\frac{9}{11}$
$\Gamma_L = 1$
$V^+ = 3V$
$I^+ = 0.06$
V=0
I=0
T
V= 3V
I= 0.06
$I^- = -0.06$
$V^- = 3V$
2T
V= 6V
I= 0
2T
$V^{++} = -2.455$
$I^{++} = -0.0491$
3T
V= 3.545
I= -0.0491
3T
$I^{--} = +0.0491$
$V^{--} = -2.455$
Time
4T
V=1.09V
I=0
4T
5T
5T


5.11V
T
2T
3T
4T
5T
6T


location, z
z=0
z=d
$\Gamma_G = -\frac{9}{11}$
$\Gamma_L = 1$
$V^+ = 3V$
$I^+ = 0.06$
V=0
I=0
T
V= 3V
I= 0.06
$I^- = -0.06$
$V^- = 3V$
2T
V= 6V
I= 0
2T
$V^{++} = -2.455$
$I^{++} = -0.0491$
3T
V= 3.545
I= -0.0491
3T
$I^{--} = +0.0491$
$V^{--} = -2.455$
Time
4T
V=1.09V
I=0
4T
$V^{+++} = 2.01V$
$I^{+++} = 0.0402$
5T
5T


location, z
z=0
z=d
$\Gamma_G = -\frac{9}{11}$
$\Gamma_L = 1$
$V^+ = 3V$
$I^+ = 0.06$
V=0
I=0
T
V= 3V
I= 0.06
$I^- = -0.06$
$V^- = 3V$
2T
V= 6V
I= 0
2T
$V^{++} = -2.455$
$I^{++} = -0.0491$
3T
V= 3.545
I= -0.0491
3T
$I^{--} = +0.0491$
$V^{--} = -2.455$
Time
4T
V=1.09V
I=0
4T
$V^{+++} = 2.01V$
$I^{+++} = 0.0402$
5T
5T
$I^{---} = -0.0402$
$V^{---} = 2.01V$
6T
V=5.11V
I=0


R2
T1
V1
5
Td=1n Z0=50
.tran 100ns
PULSE(0 3.3 0 .1ns .1ns 1s)


Tline 2 cases.raw
V(n009)
6V
5.11V
1.09V
6.0V
5.4V
4.8V
4.2V
3.6V
3.0V
2.4V
1.8V
1.2V
0.6V
0.0V
0ns
5ns
10ns
15ns
20ns
25ns
30ns
35ns
40ns
45ns
50ns

Assignment No. 1

1. Complete the bounce diagram up to 10T
2. Complete the load side (line far end) waveform diagram up to 10T
3. Draw the waveform diagram at the source side for 10T
4. Using your preferred *SPICE* simulator, simulate the model and extract voltage waveform at source and load sides (i.e. transmission line near- and far ends) and validate aforementioned waveforms.
5. Assume V_{src}= 3V, Z_0= 60Ω and T= 1^{ns}, verify waveforms of slide #6 of S01a
6. Assume T= 1^{ns}, verify waveforms of slides #6 and #9 of S01a